Masked Face Transformer

Weisong Zhao, Xiangyu Zhu, *Senior Member, IEEE*, Kaiwen Guo, Haichao Shi, *Member, IEEE*, Xiao-Yu Zhang, *Senior Member, IEEE*, and Zhen Lei, *Senior Member, IEEE*

***Abstract*— The COVID-19 pandemic makes wearing masks mandatory. Existing CNN-based face recognition (FR) systems suffer from severe performance degradation as masks occlude the vital facial regions. Recently, Vision Transformers have shown promising performance in various vision tasks with quadratic computation costs. Swin Transformer first proposes a successive window attention mechanism allowing the cross-window connection and more computational efficiency. Despite its potential, the deployment of Swin Transformer in masked face recognition encounters two challenges: 1) the attention range is insufficient to capture locally compatible face regions. 2) Masked face recognition can be defined as an occlusion-robust classification task with a known occlusion position, i.e., the position of the mask is minor-varying, which is overlooked but efficient in improving the model's recognition accuracy. To alleviate the above problem, we propose a Masked Face Transformer (MFT) with Masked Face-compatible Attention (MFA). The proposed MFA 1) introduces two additional window partition configurations, e.g., row shift and column shift, to enlarge the attention range in Swin with invariant computation costs, and 2) suppresses the interaction between the masked and non-masked regions to retain their discrepancies. Additionally, as mask occlusion leads to a separation between the masked and non-masked samples of the same identity, we propose to explore the relationship between them by a ClassFormer module to enhance intra-class aggregation. Extensive experiments show that MFT outperforms state-of-the-art masked face recognition methods in both simulated and real masked face testing datasets.**

***Index Terms*— Masked face recognition, COVID-19, face recognition, transformer.**

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Weisong Zhao, Haichao Shi, and Xiao-Yu Zhang are with the Institute of Information Engineering, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Cyber Security, University of Chinese Academy of Sciences, Beijing 101408, China (e-mail: zhaoweisong@iie.ac.cn; shihaichao@iie.ac.cn; zhangxiaoyu@iie.ac.cn).

Xiangyu Zhu and Kaiwen Guo are with the State Key Laboratory of Multimodal Artificial Intelligence Systems, Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China, and also with the School of Artificial Intelligence, University of Chinese Academy of Sciences (UCAS), Beijing 101408, China (e-mail: xiangyu.zhu@nlpr.ia.ac.cn; guokaiwen1991@gmail.com).

Zhen Lei is with the State Key Laboratory of Multimodal Artificial Intelligence Systems, Institute of Automation, Chinese Academy of Sciences, Beijing 100190, China, also with the School of Artificial Intelligence, University of Chinese Academy of Sciences (UCAS), Beijing 101408, China, and also with the CAIR, Hong Kong Institute of Science and Innovation, Chinese Academy of Sciences, Hong Kong, China (e-mail: zlei@nlpr.ia.ac.cn).

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I. Introduction

FACE recognition has achieved great success in various application domains [1], [2]. Researchers have focused on improving performance for these challenging tasks [2], [3], [4]. As the COVID-19 pandemic spreads worldwide, wearing masks has become an essential protection measure. A study by the National Institute of Standards and Technology (NIST) [5], as well as a study from the Department of Homeland Security [6], conclude that wearing masks has a significant adverse effect on the accuracy of Face Recognition systems. It is crucial to note that the masks affect not only the recognition performance of face recognition systems but also the performance of human operators [7], presentation attack detection [8], quality estimation [9], etc. Therefore, it is urgent to find technical solutions to make face recognition systems robust to wearing masks in response to the given circumstances. The importance of this topic has recently led to the organization of several competitions addressing masked face recognition [10], [11].

To improve the performance of the face recognition model against mask occlusion, a practical approach is to include labelled masked face images in the training set. However, it is quite difficult to collect a large-scale real masked face training set, as it is both labor-intensive and time-consuming. One possible solution is to convert the existing training face datasets [12], [13], [14] into simulated masked face datasets through a face masking tool [15], [16], [17]. Several approaches are proposed as part of the MFRC-21 Challenge [11], and most of them use simulated masked face datasets and train a ResNet [18] architecture with ArcFace [17] loss. The Convolution Neural Network (CNN) excels at learning local interactions between elements in the input domain, e.g., edges and contour information, while the overlapping of masks truncates their patterns. Therefore, modelling global face shape and building correlations between features in different regions are essential for masked face recognition. Recently, self-attention has been shown to effectively learn global interactions, e.g., relations between distant object parts [21], [22]. Some research has demonstrated that vision transformers perform more robustly against image occlusion than Convolution Neural Networks (CNN) [23], [24]. However, global self-attention [19] requires quadratic computation complexity. To alleviate this problem, various methods adopt the attention within regular windows to improve its efficiency [20], [25], [26], [27], [28], [29]. Swin [20] first proposes a successive attention mechanism allowing the cross-window connection. Despite its potential, the attention range cannot capture locally compatible face

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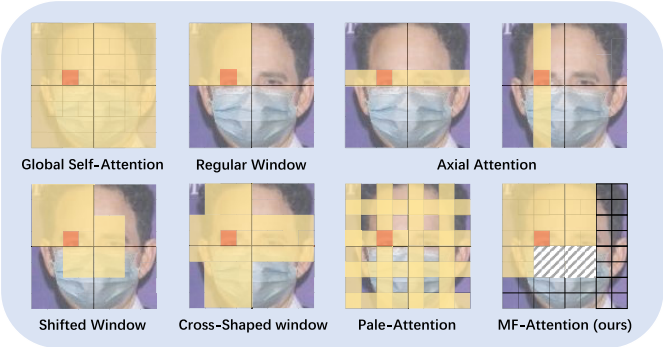

Fig. 1. An illustration of the receptive fields of different self-attention mechanisms and our MFA. Taking the left eye patch (red) as an example, global self-attention [19] possesses maximum receptive fields (yellow). Shifted Window self-attention [20] divides the input into regular windows in the current layer and shifted windows in the subsequent layer. Other methods split the input into different shapes. MFA possesses face-compatible receptive fields and restricts the information interaction between masked and non-masked face regions (grey slash) with invariant computation costs.

regions. As shown in Fig. 1, we take the eye patch as an anchor; the resulting attention range of Swin does not cover the ear and forehead regions, which is not compatible with the face topology.

Masked face recognition (MFR) can be defined as an occlusion-robust image classification task with a known occlusion position, i.e., the position of occlusion is minor-varying. Therefore, prior knowledge about occlusion positions can be incorporated to guide the model's learning. First, as specified in [30], the proportion of masked faces region is approximately 25%. Second, masks are worn on the lower half of faces and the input images are well aligned by affine transformation [17]. Both prior knowledge can be leveraged simultaneously to locate the masked face region roughly. In this paper, we locate the masked face regions in the middle of the lower face.

Given the mask occlusion position, we experimentally observe that the norm of attention between masked and non-masked regions increases as the network becomes deeper. We analyze that the model omits the discrepancies between the masked and non-masked regions at the deeper layers, which hinders the extraction of robust features against mask occlusion. Furthermore, when performing masked-to-unmasked face matching, the model shows separability between the masked and non-masked samples of the same identity [30]. Therefore, it remains challenging for deep neural networks to aggregate both masked and non-masked facial features.

To alleviate the above-mentioned problems, we propose a Masked Face Transformer (MFT) with Masked Face-compatible Attention (MFA). The proposed MFA 1) introduces two additional window partition configurations, e.g., row shift and column shift, to enlarge the attention range in Swin with invariant computation costs, and 2) suppresses the interaction between the masked and non-masked regions to retain their discrepancies. Although vanilla self-attention can effectively learn global interactions, our extensive experiments demonstrate that constraining the attention range based on the face prior further benefits the performance. Additionally, since mask occlusion leads to a separation between the masked and non-masked samples of the same identity, we propose to explore the intra-class relationship between masked and non-masked samples via the ClassFormer module, i.e., an intra-class self-attention, to enhance the intra-class aggregation.

In summary, our contributions are summarised as follows:

- We propose a Masked Face Transformer (MFT) with Masked Face-compatible Attention (MFA) to acquire a face-compatible attention range and mitigate the model forgetting in the discrepancies between masked and non-masked face regions to extract robust features.
- To the best of our knowledge, we are the first to incorporate the position prior knowledge of mask occlusion into attention design in face recognition.
- We propose to explore the relationship between intra-class masked and non-masked samples by a ClassFormer module to enhance intra-class aggregation.
- Extensive experiments on multiple synthetic and real masked face datasets demonstrate the superiority of our proposed model over the state-of-the-art methods.

II. Related Work

A. Masked Face Recognition

MFR aims to match masked and non-masked face images, which is currently a new hot and challenging task in the context of COVID-19 [6]. Considering that deep learning is a promising candidate for solving the MFR problem, large-scale masked face datasets are required for training. However, collecting a large-scale real-world masked face training set is quite difficult, as it is both labor-intensive and time-consuming. Some algorithms propose to add mask templates on normal face images to construct simulated masked face image datasets [15], [16], [31], [32]. These methods help to overcome the limited data size of masked faces. Other approaches are proposed as part of the MFRC-21 Challenge [11], and most of them use simulated masked face datasets and train a ResNet [18] architecture with ArcFace [17] loss foundation. A domain-constrained ranking loss is designed in [33] to learn two centers for normal and masked face images. To generate sufficient synthetic masked face images for training, a novel GAN with a multi-level identity preserve module is proposed in [33]. Additionally, the de-occlusion distillation framework is designed for masked face recognition [34]. Ge et al. apply GANs to reconstruct visually plausible occlusions to obtain more robust facial features against mask occlusion [35]. Zheng et al. leverage the large-scale web images without manual annotations to learn a robust representation for masked faces [36]. Qiu et al. discover corrupted features from CNNs using automatically learned masks [37]. Ding et al. [38] propose a dual-branch structure to extract both global and local features from the affine-transformed upper face region located by the Latent Part Detection (LPD) module. Zhang et al. [30] propose an attention module which extracts features from the upper half of the feature maps to assist the model's training. Huber et al. [39]

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propose a Template-level Knowledge Distillation to obtain occlusion-robust facial features. Deng et al. [40] design a masked-face images generation algorithm and propose a masked face recognition algorithm based on large margin cosine loss. Montero et al. [41] propose combining the ArcFace [18] loss with the mask-usage classification loss.

Wang et al. [42] propose alleviating attention redundancy by maximizing pairwise attention distances. In [43], pairwise self-contrastive attention-aware models are proposed to extract different local features. The proposed attention sparsity loss increases sparse responses in attention maps, thereby decreasing the focus on distracted areas. Spatial activation diversity loss and feature activation diversity loss are proposed to learn occlusion-robust face representation for the interpretable face recognition in [44], but they do not consider the prior of occlusion position. Zhao et al. [45] propose the consistent sub-decision network to synchronize the prediction of the masked faces with different quality. However, these CNN-based networks excel at learning local interactions between elements while their patterns are truncated by mask occlusion, which limits the modelling of global face shape.

B. Loss Function in Face Recognition

Recently, face recognition has achieved significant progress thanks to the great success of deep CNN models [1], [2]. Face recognition is treated as a multiclass classification problem, and deep CNN models are first introduced to learn features on large multi-identities datasets. Two types of loss functions are applied for face recognition. The first is the verification loss function. Contrastive loss [46], [47] optimizes pairwise Euclidean distance in feature space. Triplet loss [48] constructs triplets to separate the positive pair from the negative pair by a distance margin. The second is the softmax loss function, which is mostly utilized by SOTA deep face recognition methods. SphereFace [49] is proposed by introducing the angular SoftMax function, which imposes discriminative constraints on a hypersphere manifold. Moreover, the CosFace [50] with a large margin cosine loss was proposed further to maximize the decision margin in the angular space. The additive angular margin loss ArcFace loss [17] was designed to obtain highly discriminative features for FR. CurricularFace [51] embeds the idea of curriculum learning into the loss function. However, current face recognition models have a significant performance degradation when encountering masked facial images, which is urgent due to the spreading epidemic [6].

C. Visual Transformer

Transformers have recently drawn the attention of researchers in the computer vision community because of their outstanding performances in different applications. After dividing an image into non-overlapping patch tokens, Vision Transformer (ViT) [19] applies Transformer with Self-Attention to the image patches for communications among the tokens. Without delicately devised convolution kernels, ViT achieved excellent performance in image recognition. Built upon the success of ViT, many efforts have been devoted to designing better Transformer-based architectures for various vision tasks, including low-level image processing [52], image classification [19], [20], [25], [26], [29], object detection [53] and semantic segmentation [54], [55]. Some recent works [19], [20], [29] try to design a vision Transformer backbone for general-purpose vision tasks. They all follow the hierarchical Transformer architecture but adopt different self-attention mechanisms. Rather than general recognition tasks, recent studies demonstrate that the Transformer-based backbones perform more robustly against occlusion than CNNs [23], [24], which brings significant reference value for masked face recognition. Masked face recognition differs from standard occlusion recognition due to the minor-variant occlusion position and complex mask patterns. In this paper, we seek to probe the superiority of Swin Transformer [20] over CNNs (IResNet [50]) and design a more task-specified attention mechanism which is compatible with the topology of masked faces.

D. Efficient Self-Attention Mechanisms

Considering that the quadratic computation complexity is caused by self-attention, many methods are committed to improving its efficiency while avoiding performance decay [25], [26], [27]. One solution is to replace global self-attention with local self-attention, which limits the range of each self-attention layer in a regular window. As shown in the top left of Figure 1, the feature maps are first divided into several non-overlapping square regular windows, and the self-attention is performed within each window individually. Image Transformer [56] and Block-wise [57] Transformer divide a long sequence into local windows. In this case, the attention matrix has a block-diagonal structure. Only self-attention within each window is retained, and cross-bucket attention is pruned. Liu et al. [20] designate a shifted window partition to build connections across regular square windows. Besides, axial self-attention [28] achieves longer-range dependencies in horizontal and vertical directions, respectively, by performing self-attention in every single row or column of the feature map. Dong et al. [29] proposed a cross-shaped window self-attention region including multiple rows and columns. In addition to window-based local attention, stridden attention is studied widely to achieve sparse attention matrices. Sparse Transformer [58] and LongFormer [59] design the stridden attention to compute self-attention over features sampled with a sparse grid, which introduces a sparse attention matrix to provide faster computation. Swin [60] introduces the global sub-sampling layer, and Shuffle Transformer [61] introduces the shuffle module, which can also be viewed as stridden attention.

In addition to the above-mentioned local attention at the image level, several methods design the scope of local attention in the feature space. Reformer [62] splits patch tokens by feature hashing functions. Routing Transformer [63] applies online K-means to cluster tokens. BOAT [64] proposes to apply hierarchical clustering in the feature space. Sinkhorn Sorting Network [65] learns to sort and divide an input sequence into chunks. It can be seen from Fig. 1 that although the existing local attention mechanisms can provide opportunities for breaking through the local receptive fields to

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some extent, their dependencies are not rich enough to capture face topology for masked face recognition. Additionally, the existing methods overlook the key mask occlusion prior, and this paper first attempts to incorporate the position prior of mask occlusion into attention design.

E. Sample Relationship Learning

The relationship among different samples has been widely studied via various strategies [66], [67], [68]. Mini-Batch stochastic gradient descent optimization is one of the core paradigms of training deep neural networks [69]. Meanwhile, transductive inference [28,36] is also popular in few-shot approaches and recent approaches [67], [70] introduce to reason the relationships between samples, while those techniques require batch inference and thus related applications are limited. In addition, Zhang et al. [71] propose to regularize the network to favor simple linear behavior in-between training examples with mixup. In addition, Batch Normalization [72] can be considered as a core paradigm of sample relationship learning and has inspired several cross-batch methods [73], [74]. Massive domain generalization methods [75], [76], [77] aim to obtain invariant representations of the same class across domains. However, those techniques are diverse and complex, which always requires additional losses. Hou et al. [78] introduce a BatchFormer module in the penultimate layer with a shared classifier for data scarcity tasks. BatchFormerV2 is proposed by Hou et al. [79], which is pluggable into different layers in Visual Transformer and enables applying batch Transformer module for dense prediction tasks. Differently, we seek to conduct relation learning between masked and non-masked samples of the same identity to extract aggregated features.

III. The Proposed Method

As illustrated in Fig. 2, the overall architecture of the proposed Masked Face Transformer consists of MFT encoder and ClassFormer module. In this section, we first describe the proposed Masked Face-compatible Attention (MFA), including Interaction Suppression, the cyclic implementation with two new window configurations of MFA in an MFT Block, and the hierarchical pyramid structure of the encoder. Finally, we present the ClassFormer module, which consists of blended sampling and intra-class relation learning.

A. Masked Face-Compatible Attention

Swin Transformer [20] first proposes a successive attention mechanism allowing the cross-window connection. Despite its potential, the deployment of Swin Transformer in masked face recognition encounters two challenges: 1) the attention range is insufficient to capture locally compatible face regions. As shown in Fig. 3, we display the successive implementation of Swin and our method. There are four window partition configurations in our method, e.g., regular window, window shift, row shift and column shift. Specifically, we execute each attention configuration cyclically to allow the cross-window interaction, and Swin Transformer performs the former two configurations successively. Taking the left eye patch (red) as an anchor, the resulting attention range of Swin does not cover the ear and forehead regions. 2) We experimentally observe that the model omits the discrepancies between the masked and non-masked regions at the deep layers, which hinders the extraction of robust features against mask occlusion, detailed in Sec. IV-H. To tackle the above-mentioned problems, we propose Masked Face compatible Attention (MFA). Specifically, we 1) suppress the attention interaction between masked and non-masked regions in implementing MFA and 2) introduce row shift and column shift in the successive implementation of attention to fill the regions not covered.

1) Interaction Suppression: To be specific, given an input feature map $X \in \mathbb{R}^{H\times W\times C}$ (C is the channel number, H and W denote the height and width, respectively), we first split the input feature map into multiple regular-window feature maps. For each regular-window feature map $x \in \mathbb{R}^{h\times w\times c}$, we then get the query, key and value embeddings $Q, K, V \in \mathbb{R}^{h\times w\times c}$. The regular window self-attention can be formulated as:

$$Attn(Q, K, V) = \sigma(A)\cdot V,\ \ A = \frac{QK^T}{\sqrt{d_k}}, \tag{1}$$

where $\sigma(\cdot)$ indicates the Softmax function, and d_k is a constant determined by the dimension of value embedding. Let $A = \{a_{ij}\}_{i=1,2,\cdots,h\times w}^{j=1,2,\cdots,h\times w}$ indicates the set of attention values in regular-window attention maps. Since we separate a face into masked face regions and non-masked face regions with prior knowledge, the original regular window self-attention can be decoupled as follows:

$$\begin{aligned} Attn(Q, K, V) &= \sigma((\mathbb{1}_{a_{ij}\in M} + \mathbb{1}_{a_{ij}\in N} + \mathbb{1}_{a_{ij}\in C})\cdot A)\cdot V, \\ &(\mathbb{1}_{a_{ij}\in M}+\mathbb{1}_{a_{ij}\in N}+\mathbb{1}_{a_{ij}\in C}) = \mathbb{1} \in \mathbb{R}^{hw\times hw}. \end{aligned} \tag{2}$$

where M, N, C represent the attention value sets where elements are calculated from masked regions to masked regions, non-masked regions to non-masked regions, and masked regions to non-masked regions, respectively. As formulated in Eq. 2, we decouple the regular window self-attention into three parts. The coefficient matrix $(\mathbb{1}_{a_{ij}\in C}\cdot A)$ indicates the correlation between masked and non-masked face regions. Therefore, we retain the discrepancy between masked and non-masked face regions by zeroing the attention between them.

2) Row and Column Shifts: We implement the MFA by a cyclic window partition methodology including four partition configurations, e.g., regular window, window shift, and two novel configurations, e.g., row shift and column shift. As illustrated in the Fig. 3, MFA first applies a regular window partitioning strategy which starts from the upper-left pixel, and the 4×4 feature map is eventually partitioned into four windows of size 2×2 ($h, w = 2$). The second module applies the partitioning proposed in Swin Transformer [20] that displaces the windows by $(\lfloor\frac{h}{2}\rfloor, \lfloor\frac{w}{2}\rfloor)$ pixels. The third module adopts a column-shift partitioning with $(0, \lfloor\frac{w}{2}\rfloor)$ pixel displacement from the regularly partitioned windows. The final partitioning adopts a row-shift partitioning that displaces the windows by $(\lfloor\frac{h}{2}\rfloor, 0)$ pixels. From Fig. 3, we can observe

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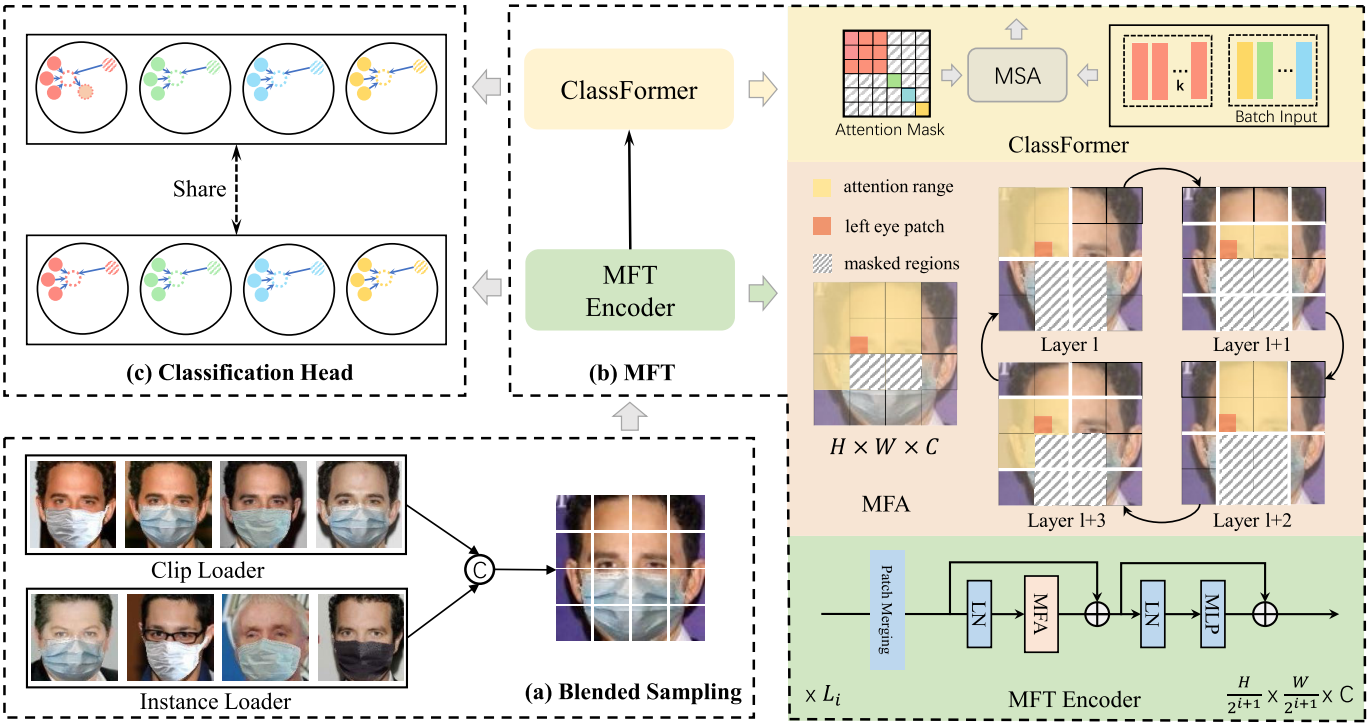

Fig. 2. The main framework of the proposed Masked Face Transformer (MFT). (a) A blended sampling strategy consisting of the instance loader and clip loader is applied to include the masked and non-mask samples of the same identity within a batch, aiming at alleviating the training oscillations. (b) The proposed MFT consists of MFT Encoder and ClassFormer. The input face images are split into multiple patches and input as tokens to the MFT encoder, where each MFT block conducts a cyclic implementation for MFA. (c) The extracted features are input into the ClassFormer module and the shared classification head to conduct relation learning between masked and non-masked samples. The hierarchical network structure is described on the right of the figure.

that Swin adopts the former two partitions, and the resulting attention range does not cover the upper forehead and middle ear regions. By contrast, the proposed row shift and column shift fill these regions. Note that all the partition configurations separate the masked and non-masked face regions and restrict attention values calculated between them by the interaction suppression described in Sec. III-A.1. With the cyclic window partition methodology, the MFT Block can be computed as follows:

$$\tilde{X}^L = \mathrm{MFA}\left(\mathrm{LN}\left(X^{L-1}\right)\right) + X^{L-1},$$
$$X^L = \mathrm{MLP}\left(\mathrm{LN}\left(\tilde{X^L}\right)\right) + \tilde{X^L}. \quad (3)$$

where $\tilde{X^L}$ and X^L denote the output of MFA and Multi-Layer Perceptron network (MLP) module in block L respectively. The choice for a partition configuration in each block depends on $L\%4$.

3) Hierarchical Pyramid Structure of MFT: The hierarchical pyramid structure consists of four stages for capturing multi-scale features by following the design in Swin [20] and CSwin [29], where each stage includes a patch merging layer. Therefore, in the i-th stage, the feature map contains $\frac{H}{2^{(i+1)}} \times \frac{W}{2^{(i+1)}}$ tokens and $2^{i-1}C$ channels. Note that the resolution of input in the last stages is equal to that of a regular window, so the attention reverts to global self-attention [19].

B. ClassFormer

1) Blended Sampling: Prior work [78] has proposed to conduct the inter-class relation learning with Multi-head Self-Attention (MSA) in a mini-batch, but we find that self-attention in a mini-batch is restricted by the long-tail or few-shot problems. By contrast, we seek to apply self-attention in the intra-class relation learning between masked and non-masked samples to enhance intra-class aggregation. An intuitive solution is to include enough samples of the same identity within each mini-batch. However, this type of sampling strategy dramatically increases the difficulty of model convergence and leads to training oscillations, as shown in Tab. V. This is due to the fact that the images sampled by the clip loader originate from the same identity, which allows

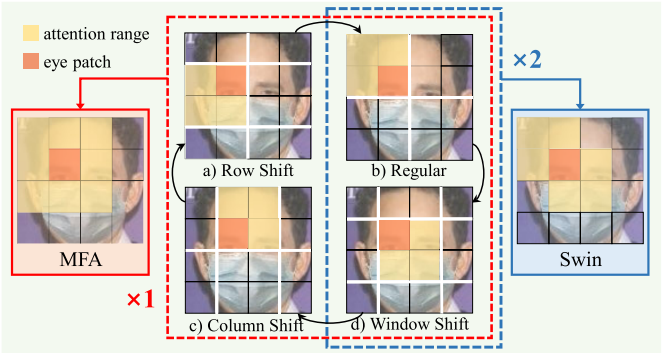

Fig. 3. An illustration of the successive implementations of Swin [20] and our method. For a clear visual comparison, we merely show the MFA without interaction suppression, and the complete implementation is shown in Fig. 2 of the manuscript. Note that Swin executes regular partition and window shift twice (×2). By contrast, we execute four window partition configurations once (×1), including a) regular window, b) window shift, c) row shift and d) column shift, which maintains the same time complexity as Swin. Taking the eye patch as the anchor patch (red), Swin adopts the former two partitions and the resulting attention range does not cover the upper forehead and middle ear regions. By contrast, row shift and column shift fill these regions.

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each mini-batch to update one class prototype. Therefore, we designate a blended sampling strategy. As shown in Fig. 2, we sample data from two data loaders, e.g., instance loader and clip loader. The clip loader is applied to obtain clip batch P by selecting t images of the same identity, i.e. $P = \{P_i\}, i = 1, 2, \cdots, t$. Moreover, we use the instance loader to sample k images of different identities as instance batch I randomly, i.e., $I = \{I_i\}, i = 1, 2, \cdots, k$, to maintain a stable execution of the training. Finally, we mix up two parts and get the mini-batch input:

$$X = I \cup P, \tag{4}$$

where $\cup$ denotes the union operation for two sets. In other words, we use a blended sampling strategy to guarantee that each input batch X consists of two sets of data, one is the set of randomly sampled images of different identities, and the other is the image set sampled from a random person. The input batch is sent to MFT Encoder to obtain the feature set, which is utilised for the following intra-class relation learning.

2) Intra-Class Relation Learning: Given the blended sampled images in a batch X, we first obtain the features by MFT-Encoder, which can be formulated as follows:

$$Z = \text{Encoder}\,(X),\ Z \in \mathbb{R}^{B\times d} \tag{5}$$

where Encoder$(\cdot)$ denotes the MFT Encoder with a hierarchical pyramid structure and multiple MFA computation, as described in Sec. III-A.2 and Sec. III-A.1. B and d indicate the batch size and feature dimension, respectively. We design the ClassFormer module with Masked Multi-head Self Attention (MMSA), Layer Normalization (LN), and Multi-Layer Perceptron (MLP) blocks to model the relationship among the masked and non-masked samples. Given the output features $Z \in \mathbb{R}^{B\times d}$ from MFT Encoder, and then we have the output of the ClassFormer as follows:

$$\begin{aligned}\tilde{Z} &= \text{LN}\,(\text{MMSA}\,(\text{Z})),\\ \hat{Z} &= \text{LN}\,(\text{MLP}\,(\tilde{Z}) + \tilde{Z}).\end{aligned} \tag{6}$$

The output feature $\hat{Z}$ from ClassFormer refers to a non-linear combination of intra-class masked and non-masked samples, which can be regarded as a virtual sample for improving the intra-class feature distribution. MMSA$(\cdot)$ indicates the Masked Multi-head Self Attention, which masks the attention values that are computed between each two samples of different identities in a batch. In other words, we maintain the intra-class attention values and omit the inter-class attention values, to improve the intra-class aggregation. Specifically, given the output features $Z \in \mathbb{R}^{B\times d}$ from MFT Encoder, we first get the query, key and value embeddings $Q, K, V \in \mathbb{R}^d$. The self-attention in a batch can be formulated as:

$$\text{MSA}(Q, K, V) = \sigma(A)\cdot V,\ \ A = \frac{QK^T}{\sqrt{d_k}}, \tag{7}$$

$\sigma(\cdot)$ indicates the Softmax function and d_k is a constant determined by the dimension of value embedding. Let $A = \{a_{ij}\}_{i=1,2,\cdots,B}^{j=1,2,\cdots,B}$ indicates the set of attention values that are computed between each two samples in a batch. B denotes the batch size. Considering whether the two samples for calculating the attention value belong to the same person, the self-attention in a mini-batch can be decoupled as follows:

$$\begin{aligned}\text{MSA}(Q, K, V) = \sigma((\mathbb{1}_{a_{ij}\in S} + \mathbb{1}_{a_{ij}\in D})\cdot A)\cdot V,\\ (\mathbb{1}_{a_{ij}\in S} + \mathbb{1}_{a_{ij}\in D}) = \mathbb{1} \in \mathbb{R}^{B\times B}.\end{aligned} \tag{8}$$

S, D represent the attention value sets where elements are calculated between two samples from the same and different identities. The coefficient matrix $(\mathbb{1}_{a_{ij}\in D}\cdot A)$ indicates the correlation between the samples of different identities. Since we seek to conduct an intra-class relation learning in a batch, we zero out the coefficient matrix $(\mathbb{1}_{a_{ij}\in D})$.

C. Shared Classification Head

We cannot perform relation learning in the test phase due to the gap between training and testing. To solve this problem, as shown in Fig. 2, we adopt a shared classifier in the training phase to learn from the features $\hat{Z}$ after ClassFormer and the features Z before ClassFormer, which is formulated as:

$$\mathcal{L}_{total} = \mathcal{L}'(\hat{Z}, W_*) + \mathcal{L}'(Z, W_*), \tag{9}$$

where W_* indicates the weights of shared classifier, and $\mathcal{L}'(\cdot,\cdot)$ denotes the CosFace classification loss [50]. Finally, we remove the ClassFormer in the test phase. The CosFace loss [50] is formulated as follows:

$$\begin{aligned}\mathcal{L}'(x_*, W_*) &= \frac{1}{N_i}\sum_i -log\frac{e^{s(cos(\theta_{y_i},i)-m)}}{e^{s(cos(\theta_{y_i},i)-m)} + \sum\limits_{j\neq y_i} e^{scos(\theta_j,i)}},\\ W &= \frac{W_*}{\|W_*\|},\ x = \frac{x_*}{\|x_*\|},\quad cos(\theta_j, i) = W_j^T x_i.\end{aligned} \tag{10}$$

where N is the number of training samples, x_i denotes the i-th feature corresponding to the ground-truth class of y_i, W_j indicates the prototype of the j-th class, θ_j refers to the angle between W_j and x_i, s is the scale hyper-parameter of model output, and m denotes the margin hyper-parameter.

IV. Experiments

A. Training Datasets

The training dataset applied in our paper is a mixture of mask and non-mask faces, which consists of two parts, i.e. (1) the first set is the MS-Celeb-1M [14], which approximately consists of 5 million non-masked face images from 93431 celebrities, and (2) the second set is generated from each image in the original dataset via FaceX-Zoo [15] from the original MS-Celeb-1M, consisting of about 5 million masked images. Specifically, we synthesize a photo-realistic masked face image using a UV texture map of the mask, which is randomly selected from eight types of masks. Fig. 5 shows the eight types of masks, including different colors and patterns, e.g., surgical mask, N95 mask, etc. The overall training dataset includes approximately 10 million non-masked face images from 93431 celebrities, and Fig. 4 shows some samples of the simulated masked face images. Therefore, the trained model can simultaneously handle two scenarios: mask vs. mask and mask vs. non-mask. The scenarios come

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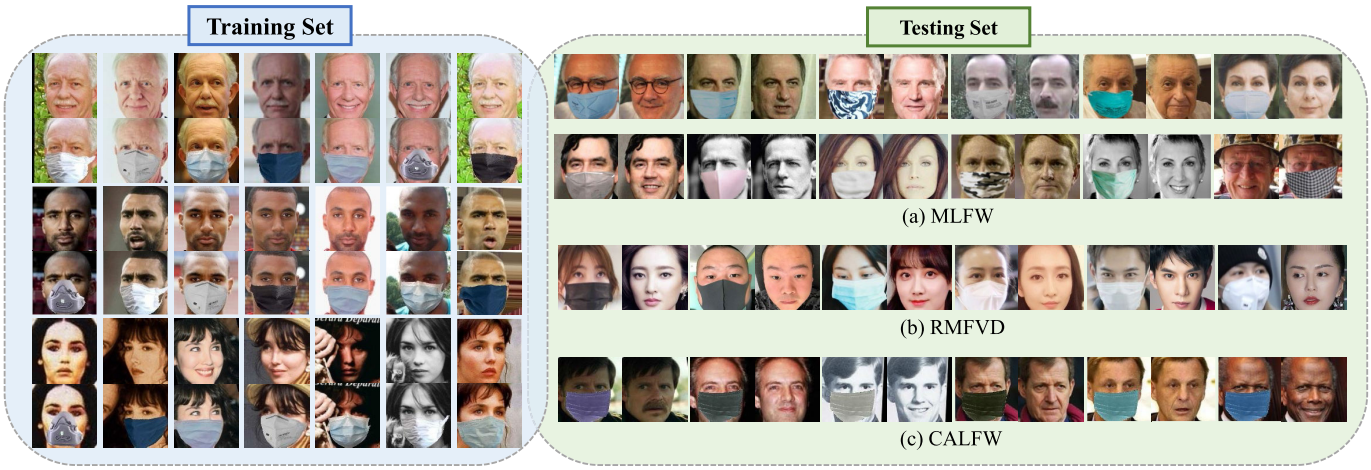

Fig. 4. Some examples of masked and normal facial images in the training and testing sets, in which the mask patterns are different, e.g., type and color.

Fig. 5. The UV map utilized for data synthesis, with different variations in types (surgical, N95, etc.) and colors (blue, white, black, etc.).

from real applications where the gallery may contain non-mask faces (by uploading) and masked faces (by snapshot), and the probe is always masked mugshot. There exists an implied requirement that the training dataset should involve masked and non-masked face images. Finally, we name the combination of the original dataset and the simulated dataset as MS-Celeb-1M (Mask).

B. Testing Datasets

a) Simulated Masked Testing Datasets: We evaluate the verification performance of our proposed method on the simulated version of three popular benchmark face datasets, including CPLFW [80], AgeDB-30 [81], and CALFW [82]. CALFW database is constructed based on the LFW [83] database, to emphasize similar-looking challenges, cross-age challenge and cross-pose challenges, and adversarial robustness of face recognition. The CPLFW database is built to facilitate large pose variation, and the AgeDB-30 database is a manually collected cross-age database. The three datasets above are converted to the simulated version by the face masking tool introduced in [15]. Following the proposed scenarios of the MFRC-21 competition [11], we evaluate both masked vs masked and masked vs non-masked comparison pairs, where we either apply the simulated mask on both images or only on one image in the comparison pair. We also evaluate our proposed method on MLFW [31], which includes 6000 pairs of masked and non-masked face images. In addition, we conduct the evaluation on LFW-masked [30], LFW-Occluded [35], and LFW-Synthetic [34], which are augmented with different mask patterns on LFW [83], for a complete demonstration on the superiority of our method.

b) Real Masked Testing Datasets: To evaluate the performance of our model on real-masked face images, we conducted experiments on MFR2 [16] and RMFVD [32]. The MFR2 consists of masked images collected in the wild of 53 identities with a total of 269 images. The authors also provide a list of 848 pairs of images to be compared. The RMFVD refers to the real masked face verification dataset, which includes 4015 masked and non-masked images of 426 identities. We clean it and construct 6000 pairs for testing.

C. Implementation Details

There are two kinds of networks applied in our work, CNNs and ViTs. For comparison, the CNN architecture used in our work is a modified ResNet-100 proposed in the first version of ArcFace [17]. As described in Section III-A.1, the ViTs architecture is a hierarchical structure that is similar to Swin Transformer [20]. In addition, we choose CosFace [40] ($s = 64$, $m = 0.4$) as the loss function for better convergence and recognition performance. The face images are aligned to 112×112 for CNNs and resized to 224×224 by bicubic interpolation to keep consistent with Swin Transformer. Different from the data augmentation applied in image recognition, we only applied the horizontal flip with a probability of 50% for both CNNs and Transformers.

We set the batch size to 128 for the CNN-based network and set the batch size to 120 for the Transformer-based network. The ratio of clip loader to instance loader is set to 1:2. We train models on 8 NVIDIA Tesla V100 (32GB) GPUs. We apply the SGD optimization method and divide the initial learning rate (0.1) at 10, 16 epochs and finish at 20 epochs for CNNs. For ViTs, we use AdamW [84] and cosine learning rate decay following DeiT [85]. The models are trained from scratch without pre-training. With four warmup epochs, the initial learning rate is set as $1e^{-3}$, and the model reaches a stable stage after ten epochs. We set weight decay to $5e^{-4}$ and $5e^{-2}$ for CNNs and ViTs, respectively.

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TABLE I

ABLATION STUDY ON THE SIMULATED AND REAL MASKED FACE VERIFICATION DATASETS INCLUDING MLFW, CPLFW-M, AgeDB-30-M AND CALFW-M, ACCURACY IN %. THE RATIO OF CLIP LOADER TO INSTANCE LOADER IS SET TO 1:2. R&C INDICATES THE ROW AND COLUMN SHIFTS IN MFA, AND IS DENOTES INTERACTION SUPPRESSION

Components					MLFW	AgeDB-30-m		CALFW-m		CPLFW-m	
Swin	Data Aug	R&C Shift	IS	ClassFormer	Both	Mask-Face	Mask-Mask	Mask-Face	Mask-Mask	Mask-Face	Mask-Mask
✓					90.36	95.11	93.33	95.11	94.05	91.03	90.08
✓	✓				94.28	97.30	97.00	95.65	95.65	92.55	92.20
✓	✓	✓			94.41	97.49	97.11	95.54	95.50	92.60	92.22
✓	✓	✓	✓		94.90	**98.10**	97.50	95.83	95.82	92.75	92.38
✓	✓	✓	✓	✓	**95.02**	98.08	**97.63**	**96.05**	**96.11**	**92.93**	**92.45**

D. Ablation Study

In the following ablation study, we investigate step by step the performance of our proposed method by looking into the (1) normal face recognition baselines of Swin, (2) mask data augmentation, (3) Row and Column Shifts (R&C Shift), (4) Interaction Suppression (IS) (5) ClssFormer. All the masked face recognition models are trained on MS-Celeb-1M (Mask). The results of different models are shown in Tab. I. The mask data augmentation improves the baseline model that is trained with normal face images. Compared to Swin with mask augmentation, the proposed row and column shifts achieve slight performance improvements; this is due to the more compatible attention range. Additionally, we find that Interaction Suppression (IS) significantly improves the performance in Mask vs. Face and Mask vs. Mask scenarios, demonstrating the proposed MFA's effectiveness. Based on this, the proposed ClassFormer further improves the model's performance by involving intra-class relation learning.

E. Comparison With the State-Of-The-Art

In this section, we compare our method with previously published models. The comparisons with SOTA are limited due to the inconsistent simulated training datasets. Therefore, we first replicate the model in MaskInv [39] as the SOTA baseline, and we also report the four models described in Section IV-D as the referenceable baselines for the MFR task. As shown in Tab. II, it can be seen that the Swin Transformer only slightly outperforms CNN with similar model parameters and computational costs. In contrast, our proposed method outperforms the MaskInv on simulated testing datasets AgeDB-30-m, CALFW-m, and CPLFW-m under both masked vs masked and masked vs non-masked scenarios. In addition, the proposed Masked Face Transformer (MFT) also outperforms the results reported in DSA [43] and HPDA [42] on MLFW and achieves a better performance than MTCosFace [40] and MFCosFace [41] on MFR2. For the real testing datasets, the proposed method achieves 1.24% improvement on RMFVD compared to Swin and achieves 99.88% on MFR2. In addition, we conduct the evaluation following [30], [34] and [45] for a complete demonstration of the superiority of our method. Due to privacy concerns, we cannot obtain the training datasets in [30], [38], and [44]. Therefore, we evaluate our model on LFW-masked to enable the comparison with them. We revise the network to Swin-Tiny [20] to maintain the consistent model size with [30], [38], [45], and [44] for fair comparisons. The results show that our method achieves the best recognition performance in all scenarios, e.g., Face-Face, Face-Mask and Mask-Mask. In addition, we apply for the occluded datasets from [34] and [35] and follow their experimental settings to train Swin-Tiny. Since Swin-Tiny is still larger than the network structure used in [34] and [35], e.g., ResNet-18, we modify the network depth of Swin-Tiny from [2,2,6,2] to [2,2,3,2] and denote it as Swin-Mini, thus achieving equal network sizes and fair comparison. As shown in Tab. IV, we replicate the results of other methods [86], [87], [88], [89], [90] from [34] and [35]. The results showcase that our method achieves superior performance on two datasets.

F. Evaluation on Normal Face Recognition

Since it is uncertain whether the detected images in real scenarios are masked, we evaluate the proposed method's performance in normal face recognition scenarios. As shown in Tab. II and Tab. III, we evaluate our model on four datasets, e.g., LFW, CALFW, CPLFW and AgeDB. Due to the shift of the distribution, the performance of our model is marginally degraded compared to the model trained with only normal face images (marked with †), but our model is ahead of other masked face recognition methods in the normal face recognition task. This is because the human eye region is shown to contain the most recognition information [91], [92]; the suppression on the lower half of the face does not affect normal face recognition performance.

G. Effects of Instance Loader and Clip Loader

As described in Section III-B, we design a blended sampling strategy consisting of intra-class sampling (Clip Loader) and inter-class sampling (Instance Loader). In this section, we conduct a study on different ratios in Tab. V. First, when the number of intra-class samples comes to zero, ClassFormer deteriorates to BatchFormer, which conducts an inter-class self-attention among randomly sampled facial images. However, we argue that the BatchFormer [78] is more compatible with long-tail or few-shot recognition tasks, while the performance on large-scale recognition tasks is inferior. From Tab. V, we can find that as the ClassFormer deteriorates to BatchFormer, the performance decreases on both simulated and real test sets. Additionally, when the number of cross-ID

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TABLE II

COMPARISON WITH SOTA ON MASKED FACE DATASETS INCLUDING CPLFW-M, AgeDB-30-M, CALFW-M, MLFW, MFR2 AND RMFVD, ACCURACY IN %. † REPRESENTS THE UPPER BOUND OF NORMAL FACE RECOGNITION PERFORMANCE. MASK-FACE & MASK-MASK INDICATES THAT THE TESTING SETS INCLUDE VERIFICATION BETWEEN MASKED FACES AND NORMAL FACES AND VERIFICATION BETWEEN TWO MASKED FACES

Methods	MLFW	RMFVD	MFR2	AgeDB-30-m			CALFW-m			CPLFW-m		
	Mask-Face & Mask-Mask			Face-Face	Mask-Face	Mask-Mask	Face-Face	Mask-Face	Mask-Mask	Face-Face	Mask-Face	Mask-Mask
CosFace [50]	90.13	90.49	98.17	98.27	94.68	92.37	96.27	94.76	93.32	93.43	89.83	87.07
CosFace152 [50]	90.47	90.64	98.52	**98.38**†	94.93	94.05	**96.37**†	94.91	94.01	93.52	91.31	88.38
Swin [20]	90.36	89.36	98.11	98.33	95.11	93.33	96.28	95.11	94.05	**93.55**†	91.03	90.08
CosFace-Mask	94.03	91.23	99.13	97.98	96.83	96.48	96.09	95.73	95.41	93.22	92.05	91.32
CosFace152-Mask	94.14	91.87	99.29	98.08	97.00	96.30	96.16	95.80	95.58	93.32	92.51	91.68
Swin-Mask [20]	94.28	91.39	99.76	98.10	97.30	97.00	95.98	95.65	95.65	93.28	92.55	92.20
MTCosFace [40]	-	-	99.41	-	-	-	-	-	-	-	-	-
MFCosFace [41]	-	-	98.50	-	-	-	-	-	-	-	-	-
DSA [43]	92.91	-	-	-	-	-	-	-	-	-	-	-
HPDA [42]	92.34	-	-	-	-	-	-	-	-	-	-	-
MaskInv [39]	93.65	91.32	99.62	97.83	96.69	96.47	96.10	95.75	95.42	92.82	91.98	91.45
From [37]	94.55	92.12	99.65	97.49	97.61	96.38	95.23	95.85	95.61	92.54	92.46	91.83
MFT (Ours)	**94.90**	**92.47**	**99.65**	98.13	**98.10**	**97.50**	96.08	**95.83**	**95.82**	93.40	**92.75**	**92.38**
+BatchFormer [78]	94.26	92.18	99.42	98.06	97.65	97.00	95.96	95.65	95.62	93.06	92.63	92.23
+BatchFormerV2 [79]	94.35	92.34	99.51	98.01	97.71	97.05	95.87	95.71	95.66	92.86	92.58	92.20
+ClassFormer (Ours)	**95.02**	**92.63**	**99.88**	98.25	**98.08**	**97.63**	96.13	**96.05**	**96.11**	93.33	**92.93**	**92.45**

TABLE III

MASKED FACE VERIFICATION RESULTS ON LFW-MASKED [30]. † REPRESENTS THE UPPER BOUND OF NORMAL FACE RECOGNITION PERFORMANCE

Method	LFW-masked		
	Face-Face	Face-Mask	Mask-Mask
ArcFace [17]	**99.84**†	97.94	98.50
ArcFace-Mask [30]	99.54	99.35	99.26
CBAM-Mask [30]	99.63	99.41	99.36
INFR [44]	99.69	97.92	97.95
LPD [38]	98.76	98.38	98.15
UPA [30]	99.58	99.41	99.37
CSN [45]	99.75	99.81	99.18
Ours	99.74	**99.84**	**99.39**

TABLE IV

MASKED FACE VERIFICATION RESULTS ON LFW-SYNTHETIC [34] AND LFW-OCCLUDED [35]

Method	Network	LFW-Synthetic	LFW-Occluded
PM [86]	ResNet-18	90.08	-
GFC [87]		95.32	91.10
CL [88]		94.43	-
CA [89]		95.97	-
GA [90]		-	91.80
ID-GAN [35]		95.63	92.32
DO [34]		-	95.44
Ours	Swin-Mini	**97.48**	**95.89**

samples comes to zero, this type of sampling strategy greatly increases the difficulty of model convergence and leads to training oscillations, which is attributed to the limited number of optimized class prototypes per batch. A more practical methodology refers to blended sampling, which includes two dataloaders. To determine the best ratio, we conduct a comparison among different ratios (1:1, 1:2, 2:1) between within-ID and cross-ID and find the ratio 1:2 achieves the best performance. Therefore, we set the ratio of clip loader to instance loader is set to 1:2 in the following comparison experiments.

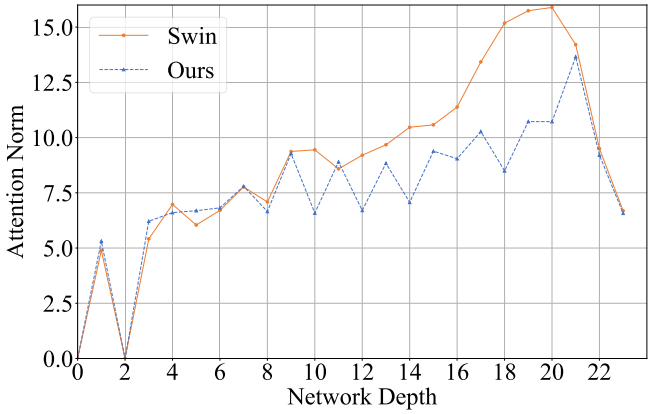

Fig. 6. Comparison of Swin with our method on the average attention norm between masked and non-masked regions at different network depths. A larger attention value indicates a closer correlation between the two regions.

H. Discrepancies Between Masked and Non-Masked Regions

We compute the average attention norm between masked and non-masked face region features at different network depths, and it can be seen from Fig. 6 that the attentions computed between masked and non-masked face regions increase as the network depth increases. This trend indicates that the model tends to rebuild the relation between masked and non-masked regions as the layer becomes deeper, which may come from the model overfitting. By contrast, we can see in the blue curve from Fig. 6 that the attention values between the discrepant regions are kept low in our model, which indicates our model maintains the discrepancies between masked and non-masked face regions by suppressing the

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TABLE V

EFFECTS OF DIFFERENT RATIOS BETWEEN WITHIN-ID SAMPLES (CLIP LOADER) AND CROSS-ID SAMPLES (INSTANCE LOADER) ON MASKED FACE VERIFICATION DATASETS. NOTE THAT ONLY SAMPLING FROM THE SAME IDENTITY IN A MINI-BATCH LEADS TO TRAINING OSCILLATIONS

Intra vs. Inter	MLFW	RMFVD	MFR2	AgeDB-30-m		CALFW-m		CPLFW-m	
	Mask-Face & Mask-Mask			Mask-Face	Mask-Mask	Mask-Face	Mask-Mask	Mask-Face	Mask-Mask
1:0	-	-	-	-	-	-	-	-	-
2:1	94.20	92.23	99.60	97.83	**97.73**	95.60	95.71	92.66	92.30
1:1	94.39	92.36	99.58	97.91	97.48	95.78	95.73	92.61	92.28
1:2	**95.02**	**92.63**	**99.88**	**98.08**	97.63	**96.05**	**96.11**	**92.93**	**92.45**
0:1 (BatchFormer [78])	94.26	92.18	99.42	97.65	97.00	95.65	95.62	92.63	92.23

TABLE VI

GENERALIZATION ON THE FACE VERIFICATION DATASETS (LFW) WITH SIMULATED SUNGLASS OCCLUSION, ACCURACY IN %

Methods	Swin	Swin-sunglass	MFT-sunglass (Ours)
LFW-sunglass	88.36	92.19	**93.16**

TABLE VII

INFERENCE SPEED COMPARISON BETWEEN CNNS, SWIN AND MFT

Methods	#Param	GFlops	Throughput (image/s)	MLFW
CosFace152-Mask	82	15.0	294	94.14
Swin-Mask	88	15.4	285	94.28
MFT-Mask	89	15.6	272	95.02

similarity between them. Moreover, the points in the Fig. 6 with an attention norm of 0 are attributed to the fact that the scale of the regular window in the shallower network is smaller than that of the masked face regions, and thus all patches within the window belong to either masked or non-masked regions.

In addition, we can observe from Fig. 6 that the Swin and our model start to decrease at the 21st layer simultaneously and achieve the same level of attention norm. The reason is: Vanilla Swin Transformer, e.g., Swin-Base, has four stages with 2, 2, 18 and 2 layers, respectively. In the layers of the last stage, the resolution of input feature maps is equal to that of a regular window. Therefore, the final attention computation mechanisms of Swin and MFT revert to global self-attention, and the output features possess high-level semantics without local information and are strongly related to the identity of the input face, resulting in the decreased trend and the final consistency of attention norm. Therefore, the attention norm in the last two layers does not reflect the relations between masked and non-masked regions but the identity information. By contrast, we find that Swin Transformer omits the difference between masked and non-masked regions at the middle and deep layers, so we retain the difference by suppressing the interaction of the two regions, and further experiments show the effectiveness of this strategy. Specifically, we conduct 500 iterations on one NVIDIA Tesla-V100 for the latency computation, and the pytorch version we installed is 1.7.1.

I. Intra-Class Aggregation

To demonstrate the improvement of our method on intra-class aggregation, we consider the intra-class cosine similarity distribution as the measurement for intra-class aggregation. Specifically, we randomly sample 5000 training samples from the training dataset to analyze the distribution of cosine similarity to the corresponding prototypes for different methods. As shown in Fig. 7a and Fig. 7b, we compare the cosine similarity distribution of CosFace, CosFace152-Mask and Swin-Mask. It can be seen that the normal face recognition model (CosFace) has two peaks since the masked faces are Out Of Distribution (OOD) samples. After training with simulated masked face datasets, the CosFace152-Mask significantly improves the intra-class cosine similarity distribution, and Swin-Mask has a slight improvement compared to CosFace152-Mask, which demonstrates the effectiveness of Transformers. By contrast, the proposed method further enhances the intra-class aggregation over Swin-Mask, as shown in Fig. 7c.

J. Generalization on Known Occlusion Position

In this section, we explore the generalization of our method to other occlusion recognition tasks, e.g., recognition with sunglass occlusion. Since masked face recognition can be defined as an occlusion-robust image classification task with a known occlusion position, i.e., the position of mask occlusion is minor-varying, we analyze that our method can generalize to the occlusion-robust image recognition problem when the occlusion region is known, e.g. face recognition with sunglass occlusion. To be specific, we first detect the landmarks to generate facial images with sunglasses from MS-Celeb-1M as the training set. For the testing set, we randomly conduct a sunglass data augmentation on half of the testing data from LFW. Then, we conduct comparisons between Swin and the proposed MFT. Considering that the occlusion position of sunglasses is different from the mask and the middle of the upper face is occluded, we revise our proposed method by replacing the occlusion region to the middle of the upper face but retaining the cyclic implementation in Section III-A.2. Tab. VI shows that our proposed method can achieve better performance in comparison to Swin-B network. Therefore, combining occlusion position detection and our proposed method could be more general for occlusion image recognition, which might be future work.

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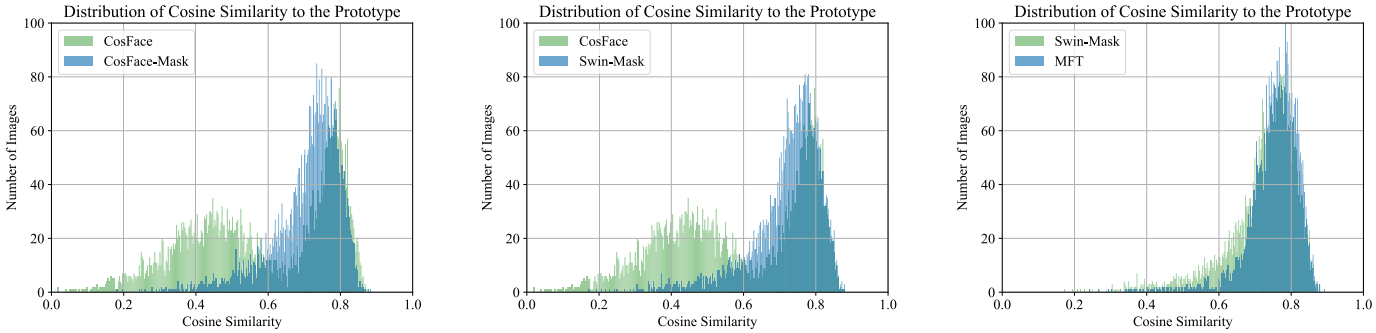

(a) CosFace vs CosFace-Mask (b) CosFace vs Swin-Mask (c) Swin-Mask vs MFT

Fig. 7. Comparison of different methods on intra-class aggregation, which is evaluated by measuring the cosine similarity from the sampled masked and non-masked images to the corresponding intra-class prototype. All the scores are intra-class similarities and it is better if they approach 1.0.

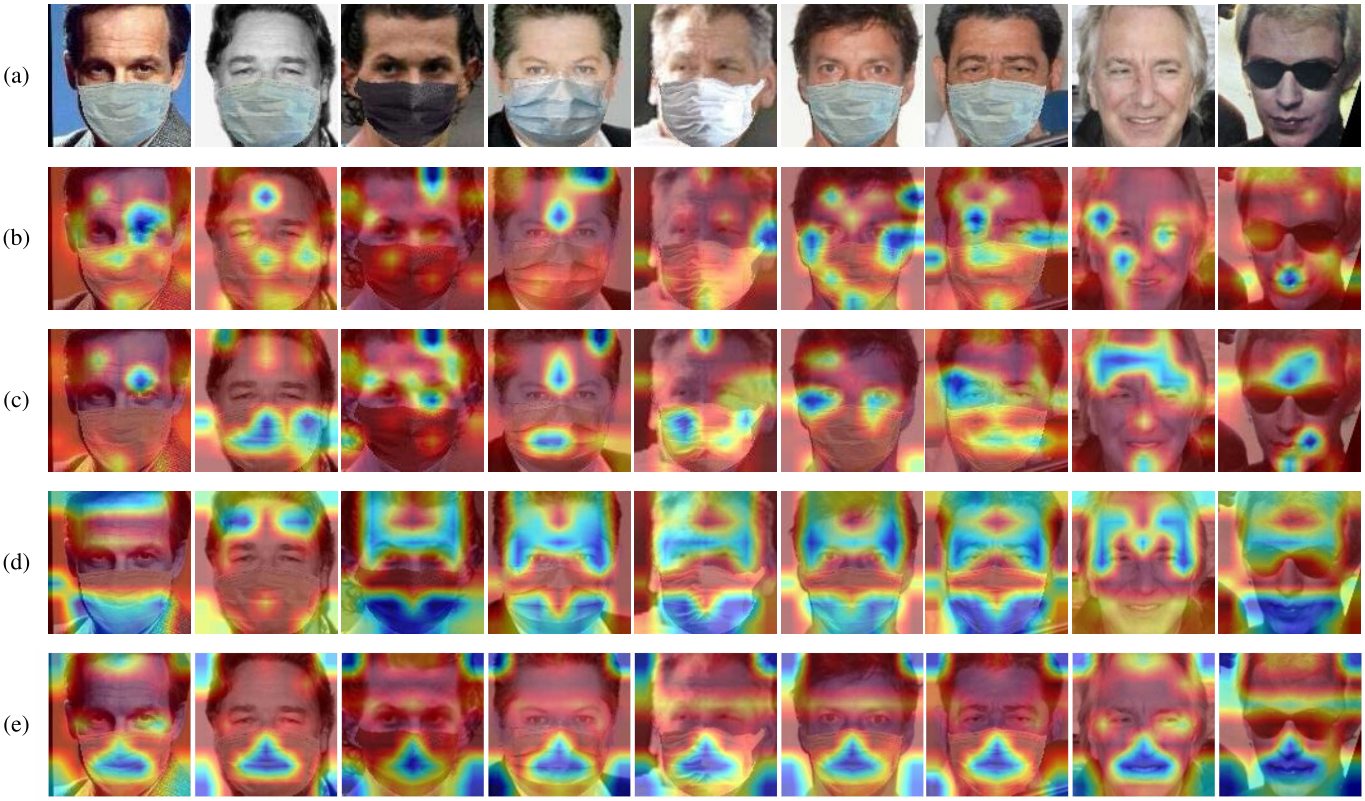

Fig. 8. ScoreCAM [93] visualization of the activation maps learned by different models. The darker heat (red) indicates that the network assigns greater weight to the corresponding regions for decision-making, and the lighter heat (blue) indicates a lower value. (a) Input masked face image, (b) CosFace, (c) CosFace152-Mask, (d) Swin-Mask, and (e) the proposed model. We can see our method pays more attention to non-masked face regions.

K. Visualization Analysis

To uncover the highly distinguishable and informative regions on masked face images focused by our proposed model, we utilize ScoreCAM [93] to visualize the activation maps of models. The visualizations of masked samples' activation maps learned by different models are shown in Fig. 8. For the CNN-based normal face recognition model, the attention is spread over the whole face image, as shown in Fig. 8b. In addition, Fig. 8c shows that the CNN-based model trained with MS-Celeb-1M (Mask) has a slight focus shift to non-masked face regions. However, all the CNN-based methods assign higher weights to the mask-occluded area. By contrast, the Swin Transformer has a clear but irrelevant focus on both masked and non-masked face regions, as shown in Fig. 8d. In comparison to other models, our model pays more attention to non-masked face regions for masked samples, as shown in Fig. 8e. For non-masked samples, our method produces a similar activation heatmap with the masked facial images, i.e. the heat of middle-lower face regions is higher than that of other regions, which demonstrates that our method can guide the model to focus on the non-masked facial regions. For samples with upper-face occlusion, the heat is similar with both masked and non-masked facial images, but the region of upper-face occlusion is activated with a high heat value, which is not conducive for robust feature extraction against upper-face occlusion. Moreover, we find that our method suppresses focus on the masked region for both masked face images and normal face images, which is beneficial for Mask-Mask vs. Mask-Face scenes. For Face-Face scenarios, since the human eye region is shown to contain the most recognition information [91], [92], the suppression

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(a) Upper-face Occlusion (b) Age Shift

Fig. 9. Some positive image pairs of the same identity that are not matched by the model correctly, which involves (a) upper-face occlusion and (b) age shift.

on the lower half of the face does not affect normal face recognition performance, which is also confirmed by the results in Tab. II and Tab. III.

L. Inference Speed Comparison

To have an explicit comparison of the model inference speed between CNN-based methods, i.e. IR152, Swin and our proposed method, we calculate the model size (#param and GFlops) and inference speed (throughput) of IR152, Swin and MFT (ours), respectively. Note that we remove the weights of the classification head and stop GFlops computation after extracting the feature. From the following Tab. VII, we can find that the proposed MFT achieves a significant improvement in comparison to the CNN-based network IR152 on the MLFW test set, which has similar Flops and model size. Additionally, the throughputs of IR152 and our MFT are also equivalent, which is implemented on a V100 GPU. By contrast, our approach also maintains comparable GFlops and parameter size compared to Swin. This is because (1) our hierarchical network structure is consistent with Swin and (2) the proposed Row and Column shifts replace the window shift in Swin, which does not significantly increase computational cost. The marginal variance in parameter size and inference speed is due to the additional mask to conduct interaction suppression and the roll manipulation in Pytorch implementation, i.e., $torch.roll()$, respectively. In addition, the proposed ClassFormer module is discarded in the final test phase, which maintains invariant computation costs.

V. Dicussion

In this part, we discuss the limitation of the proposed method. Specifically, the proposed Masked Face Transformer can focus on face regions without mask occlusion as illustrated in Section IV-K. However, this concentration also leads to the inability to handle images with upper-face occlusions like glasses and hats, as shown in Fig. 9a, and Section IV-J provides a referenceable solution, which contains a minor modification to the MFT, to tackle the problem caused by upper-face occlusion with known occlusion position. However, the proposed method cannot bring improvements to the occlusion recognition tasks with unknown occlusion locations due to the prior-based attention design. In addition, Fig. 9b shows some failure cases with large age variation which is a fatal problem for face recognition. Although our method cannot perform well on upper-face occlusion recognition, our method can be slightly revised to fit novel occlusion scenarios as long as the occlusion position is given.

VI. Conclusion

To obtain face-compatible cross-window attention with invariant computation costs while tentatively leveraging the prior knowledge of mask occlusion, we first locate the mask occlusion region in the middle of the lower face and propose to incorporate mask occlusion prior knowledge into attention design. In this paper, we present a Masked Face Transformer with Masked Face-compatible Attention which not only obtains a face-compatible range of local attention with invariant computation complexity, but also mitigates the model's oblivion in the discrepancies between masked and non-masked face regions. Additionally, considering that mask occlusion leads to a separation between the masked and non-masked samples of the same identity, we propose to explore the relationship between the masked and non-masked samples and present a ClassFormer module, which refers to an intra-class self-attention, to enhance the intra-class aggregation further. Extensive experiments in a detailed ablation study and comparison with SOTA approaches demonstrate the effectiveness and superiority of our method over existing methods. Additionally, further experiments illustrate that the proposed method pays more attention to non-masked face regions for masked samples and effectively suppresses the degradation of the discrepancy between masked and non-masked regions. Moreover, experiments on sunglass occlusion recognition demonstrate the generalization of the proposed method, which is not only applicable to the masked face recognition task, but also has reference significance for other recognition problems with known occlusion position. Masked face recognition, which has been brought into focus by the global COVID-19 pandemic, will be of further interest to our society and require novel solutions. Additionally, the potential of the presented Masked Face Transformer may not be limited to its use for masked face recognition but could also provide a significant reference for the problem of known occlusion recognition in general, which is still to be studied.

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Weisong Zhao received the B.E. degree in computer science and technology from Henan University in 2021. He is currently pursuing the joint Ph.D. degree in computer application technology with the School of Cyber Security, University of Chinese Academy of Sciences, and the Institute of Information Engineering, Chinese Academy of Sciences, Beijing, China. His current research interests include biometrics, pattern recognition, and computer vision.

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Xiangyu Zhu (Senior Member, IEEE) received the B.S. degree from Sichuan University (SCU) in 2012 and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences, in 2017. He is currently an Associate Professor with the Chinese Academy of Sciences. He has published 50 articles, including top journals, such as IEEE TRANSACTIONS ON PATTERN ANALYSIS AND MACHINE INTELLIGENCE and *International Journal of Computer Vision*, and top vision conferences, such as ICCV, CVPR, and ECCV. His total citation in Google Scholar is more than 5,000 times. His current research interests include pattern recognition and computer vision, particularly 3D face models, face anti-spoofing, and face recognition. He has won the 2019 Best Doctoral Dissertation Finalist in CSIG, the Champion of the ICCV19 Light Weight Face Recognition Challenge, and the CCBR Best Paper Award. He served as the Area Chair for the ACM Multimedia (MM) in 2020 and 2021.

Kaiwen Guo received the B.S. degree from the Changchun University of Technology in 2017. His current research interests include computer vision and machine learning with applications to face recognition and facial analysis, 3-D modeling from video, image, and video-based recognition and exploitation.

Haichao Shi (Member, IEEE) received the B.S. degree in software engineering from Beijing Technology and Business University, Beijing, China, in 2017, and the Ph.D. degree in cyberspace security from the Institute of Information Engineering, Chinese Academy of Sciences, Beijing, in 2022. He is currently an Engineer with the Institute of Information Engineering, Chinese Academy of Sciences. His current research interests include temporal action localization, action recognition, and pattern recognition.

Xiao-Yu Zhang (Senior Member, IEEE) received the B.S. degree in computer science from the Nanjing University of Science and Technology, Nanjing, China, in 2005, and the Ph.D. degree in pattern recognition and intelligent systems from the Institute of Automation, Chinese Academy of Sciences, Beijing, China, in 2010. He is currently an Associate Professor with the Institute of Information Engineering, Chinese Academy of Sciences. He has authored or coauthored more than 60 refereed publications in international journals and conferences. His current research interests include artificial intelligence, data mining, and computer vision. He is a Senior Member of ACM, CCF, and CSIG. His awards and honors include the Silver Prize in the Microsoft Cup of the IEEE China Student Paper Contest in 2009, the Second Prize in the Wu Wen-Jun AI Science and Technology Innovation Award in 2016, the CCCV Best Paper Nominate Award in 2017, the Third Prize of BAST Beijing Excellent S&T Paper Award in 2018, and the Second Prize of CSIG Science and Technology Award in 2019.

Zhen Lei (Senior Member, IEEE) received the B.S. degree in automation from the University of Science and Technology of China in 2005, and the Ph.D. degree from the Institute of Automation, Chinese Academy of Sciences, in 2010. He is currently a Professor with the Institute of Automation, Chinese Academy of Sciences. He has published more than 200 papers in international journals and conferences with more than 27,000 citations in Google Scholar and an H-index of 80. His current research interests include computer vision, pattern recognition, image processing, and face recognition. He is an IAPR Fellow and an AAIA Fellow. He is the Winner of the 2019 IAPR Young Biometrics Investigator Award. He is the Program Co-Chair of IJCB in 2023, the Competition Co-Chair of IJCB in 2022, and the area chair of several conferences. He is an Associate Editor of IEEE TRANSACTIONS ON INFORMATION FORENSICS AND SECURITY, *Pattern Recognition*, *Neurocomputing*, and *IET Computer Vision*.

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